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Microscopic dynamics of s sbar -> Lambda Lambda-bar spin correlations

The goal is to construct a microscopic dynamical description that starts from an initially produced s bar s pair, evolves its two-spin density matrix through coherent and dissipative QCD dynamics, propagates the spin information through hadronization and feed-down, and predicts the final Lambda bar Lambda spin-correlation observable. The principal result obtained on 22 August is a selection rule: a broad class of exchange-symmetric coherent two-spin Hamiltonians preserves the scalar quantity langle bm sigma_1 ! cdot ! bm sigma_2 rangle. Therefore the observed decrease of the STAR scalar correlation cannot be explained by common coherent precession or ordinary symmetric exchange interactions alone. The dynamical problem is sharpened to the calculation of singlet--triplet conversion rates generated by relative spin fields, exchange-antisymmetric couplings, or genuinely collisional/dissipative terms. This report derives that statement step by step and explains how it fits into the emerging s bar s to Lambda bar Lambda framework.

OBSERVABLE AND DENSITY-MATRIX CONVENTIONS

For two spin-1/2 particles we write
rho= frac14 left[Id otimes Id+P_i sigma_i otimes Id+ bar P_j Id otimes sigma_j+C_ij sigma_i otimes sigma_j right].
The tensor C_ij contains the genuine two-spin information,
C_ij= Trc ! left[rho , sigma_i otimes sigma_j right].
The scalar relative-polarization observable is
boxedP= frac13 Trc C= frac13 left langle bm sigma_1 cdot bm sigma_2 right rangle.
STAR reports a positive short-range Lambda bar Lambda value around
P_ Lambda bar Lambda=0.181 pm0.035_ rm stat pm0.022_ rm sys,
with a substantial reduction at larger pair separation. The theory must explain both the positive short-distance source and the subsequent reduction.

INITIAL ${}^3P_0$ SOURCE

A vacuum-like 0^++ pair-creation mechanism implies
L=1, qquad S=1, qquad J=0.
The coupled state is
|00 rangle= frac1 sqrt3 sum_m=-1^1(-1)^1-m|1m rangle_L|1,-m rangle_S.
If the orbital projection is not observed, tracing over the L=1 sector gives
rho_S&= Trc_L|00 rangle langle00|
&= frac13 sum_m=-1^1|1m rangle langle1m|
&= frac13P_S=1.
For two spin-1/2 degrees of freedom,
P_S=1= frac3 Id+ bm sigma_1 cdot bm sigma_24,
therefore
rho_S= frac14 left[Id otimes Id+ frac13 sum_i sigma_i otimes sigma_i right],
and hence

C_ij^(0)= frac13 delta_ij, qquad P^(0)= frac13.
This result is the orbitally averaged spin state. A momentum-conditioned state can retain additional anisotropic information, so an event-by-event theory should postpone orbital averaging until the experimental projection is specified.

EXACT COHERENT EVOLUTION OF THE FULL TENSOR

Consider the most general real bilinear spin Hamiltonian
H= frac12 Omega_a sigma_a otimes Id+ frac12 bar Omega_b Id otimes sigma_b+ frac14J_ab sigma_a otimes sigma_b.
The density matrix obeys the exact von Neumann equation
dot rho=-i[H, rho].
Differentiating C_ij gives
dot C_ij
&= Trc ! left[dot rho , sigma_i otimes sigma_j right]
&=-i Trc ! left([H, rho] , sigma_i otimes sigma_j right)
&=-i left langle[sigma_i otimes sigma_j,H] right rangle,
where cyclicity of the trace was used in the last line.
For the first local field,
-i left[sigma_i otimes sigma_j, frac12 Omega_a sigma_a otimes Id right]
&=-i frac Omega_a2[sigma_i, sigma_a] otimes sigma_j
&= Omega_a epsilon_iak sigma_k otimes sigma_j,
so
left. dot C_ij right|_1= epsilon_iak Omega_aC_kj.
Likewise,
left. dot C_ij right|_2= epsilon_jbl bar Omega_bC_il.
For the interaction term we use
[A otimes B,C otimes D]=[A,C] otimes frac B,D 2+ frac A,C 2 otimes[B,D],
together with
[sigma_i, sigma_a]=2i epsilon_iak sigma_k, qquad sigma_i, sigma_a =2 delta_ia Id.
Thus
[sigma_i otimes sigma_j, sigma_a otimes sigma_b]
&=2i epsilon_iak delta_jb sigma_k otimes Id
& quad+2i delta_ia epsilon_jbl Id otimes sigma_l.
After multiplication by J_ab/4,
left. dot C_ij right|_12= frac12 epsilon_iakJ_ajP_k+ frac12 epsilon_jblJ_ib bar P_l.
Combining all terms yields
boxed beginaligned
dot C_ij=& epsilon_iak Omega_aC_kj+ epsilon_jbl bar Omega_bC_il
&+ frac12 epsilon_iakJ_ajP_k+ frac12 epsilon_jblJ_ib bar P_l.
This equation is exact for coherent bilinear dynamics and contains no Markov or weak-coupling approximation.

PROTECTION THEOREM FOR THE STAR SCALAR

Define
Q equiv bm sigma_1 cdot bm sigma_2.
Since 3P= langle Q rangle, determining whether P changes is equivalent to determining whether Q is conserved. The two-spin swap operator is
mathsf P_12= frac12(Id+Q),

so
Q=2 mathsf P_12- Id.
If the Hamiltonian is invariant under exchange,
mathsf P_12H mathsf P_12=H,
then
[H, mathsf P_12]=0 quad Longrightarrow quad[H,Q]=0.
Therefore
boxed fracddt langle Q rangle=0, qquad fracddt Trc C=0.
For the Hamiltonian above, exchange invariance requires
boxed bm Omega= bar bm Omega, qquad J_ij=J_ji.
Hence common spin precession and any exchange-symmetric bilinear interaction can rotate or redistribute
components of C_ij but cannot alter the scalar P.
In particular, ordinary Heisenberg exchange
H_ rm ex=J(r) , bm S_1 cdot bm S_2
is incapable of producing the observed scalar suppression by itself.

REFORMULATION IN TERMS OF SINGLET PROBABILITY

The singlet projector is
Pi_S=|S rangle langle S|= frac14 left(Id- bm sigma_1 cdot bm sigma_2 right).
Therefore
p_S&= Trc(rho Pi_S)
&= frac14 left[1- langle Q rangle right]
&= boxed frac1-3P4.
For an initially pure triplet sector, p_S(0)=0 and P(0)=1/3. A decrease of P therefore means that singlet weight has been generated. At the level of the measured STAR central value,
p_S^ rm obs= frac1-3(0.181)4 simeq0.114,
although this number cannot yet be interpreted as a primordial partonic singlet fraction because
hadronization and feed-down have not been unfolded.
The microscopic question is consequently more precise than ‘‘what causes decoherence?’’:
boxed textWhat QCD dynamics transfers probability between S=1 text and S=0?

INTERACTIONS CAPABLE OF SINGLET--TRIPLET CONVERSION

Write the local fields as common and relative pieces,
bm Omega_c= frac bm Omega+ bar bm Omega2, qquad
delta bm Omega= frac bm Omega- bar bm Omega2.
Then
H_ rm local= bm Omega_c cdot(bm S_1+ bm S_2)+ delta bm Omega cdot(bm S_1- bm S_2).
The first term is exchange symmetric; the second is exchange antisymmetric.
For a relative field along z,
H_ delta= frac delta Omega_z2(sigma_1z- sigma_2z).
Using
|T_0 rangle= frac| uparrow downarrow rangle+| downarrow uparrow rangle sqrt2, qquad
|S rangle= frac| uparrow downarrow rangle-| downarrow uparrow rangle sqrt2,
we obtain
(sigma_1z- sigma_2z)|T_0 rangle=2|S rangle.
Thus a differential field directly mixes the triplet and singlet sectors. Starting from |T_0 rangle,
p_S(t)= sin^2(delta Omega_z t),
and therefore

boxedP(t)= frac13 left[1-4 sin^2(delta Omega_z t) right].
A second coherent possibility comes from the antisymmetric part of J_ij,
J^(A)_ij= frac12(J_ij-J_ji)= epsilon_ijkD_k.
The corresponding interaction is
H_A= bm D cdot(bm S_1 times bm S_2),
which is odd under exchange and can also couple singlet and triplet sectors.

KINETIC EQUATION SUGGESTED BY THE COHERENT ANALYSIS

The scalar problem can be coarse-grained into a rate equation for p_S,
boxed dot p_S= Gamma_T to Sp_T- Gamma_S to Tp_S+S_S^ rm prod,
with p_T=1-p_S when no probability leaks outside the two-spin subspace. Since
P= frac1-4p_S3,
we have
dot P=- frac43 dot p_S.
The central microscopic quantities are therefore the conversion rates
Gamma_T to S, qquad Gamma_S to T,
not an unconstrained phenomenological damping coefficient.

CONNECTION TO THE WIGNER/KADANOFF--BAYM ROUTE

For a massive quark, the gauge-covariant Wigner function may be decomposed as
W= frac14 left[mathcal F+i gamma^5 mathcal P+ gamma^ mu mathcal V_ mu+ gamma^ mu gamma^5
mathcal A_ mu+ frac12 sigma^ mu nu mathcal S_ mu nu right].
The axial-vector component mathcal A^ mu carries spin information. Existing relativistic
quantum
kinetic theory derives one-particle transport equations for mathcal A^ mu containing
color-field
correlators and spin-diffusion/source terms. Our problem requires the two-particle object
W^(2)(1,2)= left langle widehat W_s(1) otimes widehat W_ bar s(2) right rangle,
and especially the connected axial--axial correlator
mathcal C^ mu nu(1,2)= left langle mathcal A_s^ mu(1) mathcal A_ bar s^ nu(2) right
rangle_c.
After a two-particle Kadanoff--Baym derivation, the collision kernel should be projected
onto
Pi_S= frac14(Id- bm sigma_1 cdot bm sigma_2), qquad Pi_T= Id- Pi_S.
The target definitions are schematically
boxed Gamma_T to S sim Trc ! left[Pi_S mathcal C_ rm coll[Pi_T] right], qquad
Gamma_S to T sim Trc ! left[Pi_T mathcal C_ rm coll[Pi_S] right].
This is the next route toward a microscopic QCD calculation.

HADRONIZATION AND FEED-DOWN

The partonic tensor is not measured directly. A channel-resolved map is required,
C_ij^ Lambda bar Lambda= sum_abR_ab(A_a)_ikC_kl^s bar s,(ab)(bar A_b)_jl,
where a,b label primary and feed-down channels and A_a, bar A_b are their spin-transfer
matrices.
Different channels need not have the same transfer coefficient, so a single scalar feed-down
dilution
is not generally sufficient.

CURRENT INTERPRETATION AND NEXT STEPS

The robust chain emerging from the analysis is
boxed textcorrelated s bar s text source to textsinglet--triplet transport to
textchannel-resolved hadronization/feed-down to Lambda bar Lambda.
The highest-value next tasks are:
- derive the two-particle axial--axial Kadanoff--Baym equation with explicit color factors;
- identify which QCD self-energy structures generate relative-spin or exchange-antisymmetric kernels;
- determine whether confinement produces coherent oscillatory conversion, irreversible conversion, or both;
- construct explicit spin-transfer matrices for primary Λ, Σ^0, Ξ, and mixed pairs;
- eventually fit tensor observables $C_T\(\Delta R\)$ and $C_L\(\Delta R\)$ rather than only their trace.

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